In [1]:
```
!pip install yfinance
```

```
Collecting yfinance
  Downloading yfinance-1.2.2-py2.py3-none-any.whl.metadata (6.1 kB)
Requirement already satisfied: pandas>=1.3.0 in d:\ana\lib\site-packages (from yfinance) (2.3.3)
Requirement already satisfied: numpy>=1.16.5 in d:\ana\lib\site-packages (from yfinance) (2.3.5)
Requirement already satisfied: requests>=2.31 in d:\ana\lib\site-packages (from yfinance) (2.32.5)
Collecting multitasking>=0.0.7 (from yfinance)
  Using cached multitasking-0.0.12-py3-none-any.whl
Requirement already satisfied: platformdirs>=2.0.0 in d:\ana\lib\site-packages (from yfinance) (4.5.0)
Requirement already satisfied: pytz>=2022.5 in d:\ana\lib\site-packages (from yfinance) (2025.2)
Requirement already satisfied: frozendict>=2.3.4 in d:\ana\lib\site-packages (from yfinance) (2.4.6)
Collecting peewee>=3.16.2 (from yfinance)
  Downloading peewee-4.0.4-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: beautifulsoup4>=4.11.1 in d:\ana\lib\site-packages (from yfinance) (4.13.5)
Collecting curl_cffi>=0.15 (from yfinance)
  Downloading curl_cffi-0.15.0-cp310-abi3-win_amd64.whl.metadata (18 kB)
Requirement already satisfied: protobuf>=3.19.0 in d:\ana\lib\site-packages (from yfinance) (5.29.3)
Collecting websockets>=13.0 (from yfinance)
  Using cached websockets-16.0-cp313-cp313-win_amd64.whl.metadata (7.0 kB)
Requirement already satisfied: soupsieve>1.2 in d:\ana\lib\site-packages (from beautifulsoup4>=4.11.1->yfinance) (2.5)
Requirement already satisfied: typing-extensions>=4.0.0 in d:\ana\lib\site-packages (from beautifulsoup4>=4.11.1->yfinance)
(4.15.0)
Requirement already satisfied: cffi>=2.0.0 in d:\ana\lib\site-packages (from curl_cffi>=0.15->yfinance) (2.0.0)
Requirement already satisfied: certifi>=2024.2.2 in d:\ana\lib\site-packages (from curl_cffi>=0.15->yfinance) (2025.11.12)
Requirement already satisfied: rich in d:\ana\lib\site-packages (from curl_cffi>=0.15->yfinance) (14.2.0)
Requirement already satisfied: pycparser in d:\ana\lib\site-packages (from cffi>=2.0.0->curl_cffi>=0.15->yfinance) (2.23)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\ana\lib\site-packages (from pandas>=1.3.0->yfinance) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in d:\ana\lib\site-packages (from pandas>=1.3.0->yfinance) (2025.2)
Requirement already satisfied: six>=1.5 in d:\ana\lib\site-packages (from python-dateutil>=2.8.2->pandas>=1.3.0->yfinance) (1.1
7.0)
Requirement already satisfied: charset_normalizer<4,>=2 in d:\ana\lib\site-packages (from requests>=2.31->yfinance) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in d:\ana\lib\site-packages (from requests>=2.31->yfinance) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in d:\ana\lib\site-packages (from requests>=2.31->yfinance) (2.5.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in d:\ana\lib\site-packages (from rich->curl_cffi>=0.15->yfinance) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in d:\ana\lib\site-packages (from rich->curl_cffi>=0.15->yfinance) (2.1
9.2)
Requirement already satisfied: mdurl~=0.1 in d:\ana\lib\site-packages (from markdown-it-py>=2.2.0->rich->curl_cffi>=0.15->yfina
nce) (0.1.2)
Downloading yfinance-1.2.2-py2.py3-none-any.whl (130 kB)
Downloading curl_cffi-0.15.0-cp310-abi3-win_amd64.whl (1.7 MB)
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   ---------------------------------------- 1.7/1.7 MB 735.0 kB/s  0:00:02
Downloading peewee-4.0.4-py3-none-any.whl (144 kB)
Using cached websockets-16.0-cp313-cp313-win_amd64.whl (178 kB)
Installing collected packages: peewee, multitasking, websockets, curl_cffi, yfinance

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  ---------------------------------------- 5/5 [yfinance]

Successfully installed curl_cffi-0.15.0 multitasking-0.0.12 peewee-4.0.4 websockets-16.0 yfinance-1.2.2
```

In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import yfinance as yf
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings("ignore")

sns.set_style("whitegrid")
print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

In [2]:
```python
# Download Apple stock data from Yahoo Finance
ticker = "AAPL"
print(f"Downloading {ticker} historical data (2020-2024)...")

df = yf.download(ticker, start="2020-01-01", end="2024-12-31", progress=False)
df.columns = [col[0] if isinstance(col, tuple) else col for col in df.columns]
df.reset_index(inplace=True)

print(f"Data downloaded! Shape: {df.shape}")
print(f"Date range: {df['Date'].min().date()} to {df['Date'].max().date()}")
print()
print("First 5 rows:")
df.head()
```

```
Downloading AAPL historical data (2020-2024)...
Data downloaded! Shape: (1257, 6)
Date range: 2020-01-02 to 2024-12-30

First 5 rows:
```

Out[2]:

	Date	Close	High	Low	Open	Volume
0	2020-01-02	72.400505	72.460769	71.156667	71.409770	135480400
1	2020-01-03	71.696632	72.455950	71.472454	71.629138	146322800
2	2020-01-06	72.267921	72.306491	70.568495	70.819193	118387200
3	2020-01-07	71.928055	72.533095	71.708695	72.277578	108872000
4	2020-01-08	73.085098	73.386416	71.631544	71.631544	132079200

In [3]:

```
# Drop rows with missing values
df.dropna(inplace=True)

# Feature Engineering
df["MA_5"]            = df["Close"].rolling(window=5).mean()
df["MA_10"]           = df["Close"].rolling(window=10).mean()
df["Price_Change"]    = df["Close"].pct_change()
df["High_Low_Range"] = df["High"] - df["Low"]
df["Target"]          = df["Close"].shift(-1)   # Next day close

df.dropna(inplace=True)
df.reset_index(drop=True, inplace=True)

print("Features after engineering:")
print(df.columns.tolist())
print(f"\nFinal dataset shape: {df.shape}")
df[["Date","Open","High","Low","Close","Volume","MA_5","Target"]].tail(5)
```

```
Features after engineering:
['Date', 'Close', 'High', 'Low', 'Open', 'Volume', 'MA_5', 'MA_10', 'Price_Change', 'High_Low_Range', 'Target']

Final dataset shape: (1247, 11)
```

Out[3]:

	Date	Open	High	Low	Close	Volume	MA_5	Target
1242	2024-12-20	246.692406	253.614600	244.355183	253.107376	147495300	250.004300	253.883133
1243	2024-12-23	253.385849	254.261058	252.073014	253.883133	40858800	250.845709	256.797211
1244	2024-12-24	254.101927	256.807136	253.903002	256.797211	23234700	251.784586	257.612701
1245	2024-12-26	256.787224	258.686851	256.230269	257.612701	27237100	253.966660	254.201370
1246	2024-12-27	256.429191	257.294489	251.685117	254.201370	42355300	255.120358	250.829803

In [4]:

```
feature_cols = ["Open", "High", "Low", "Volume", "MA_5", "MA_10",
                "Price_Change", "High_Low_Range"]

X = df[feature_cols].values
y = df["Target"].values

# Time-series split — no shuffle!
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, shuffle=False
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled  = scaler.transform(X_test)

print(f"Training samples : {len(X_train)}")
print(f"Testing  samples : {len(X_test)}")
print(f"Features used    : {feature_cols}")
```

```
Training samples : 997
Testing  samples : 250
Features used    : ['Open', 'High', 'Low', 'Volume', 'MA_5', 'MA_10', 'Price_Change', 'High_Low_Range']
```

In [5]:

```
# Model 1: Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)
lr_preds = lr_model.predict(X_test_scaled)

# Model 2: Random Forest
```

```python
rf_model = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf_model.fit(X_train_scaled, y_train)
rf_preds = rf_model.predict(X_test_scaled)

print("Both models trained!")
print()
print("LINEAR REGRESSION RESULTS:")
print(f"  MAE  : ${mean_absolute_error(y_test, lr_preds):.2f}")
print(f"  RMSE : ${np.sqrt(mean_squared_error(y_test, lr_preds)):.2f}")
print(f"  R2   :  {r2_score(y_test, lr_preds):.4f}")
print()
print("RANDOM FOREST RESULTS:")
print(f"  MAE  : ${mean_absolute_error(y_test, rf_preds):.2f}")
print(f"  RMSE : ${np.sqrt(mean_squared_error(y_test, rf_preds)):.2f}")
print(f"  R2   :  {r2_score(y_test, rf_preds):.4f}")
```

```
Both models trained!

LINEAR REGRESSION RESULTS:
  MAE  : $2.29
  RMSE : $3.04
  R2   :  0.9857

RANDOM FOREST RESULTS:
  MAE  : $19.12
  RMSE : $25.88
  R2   :  -0.0346
```

In [6]:

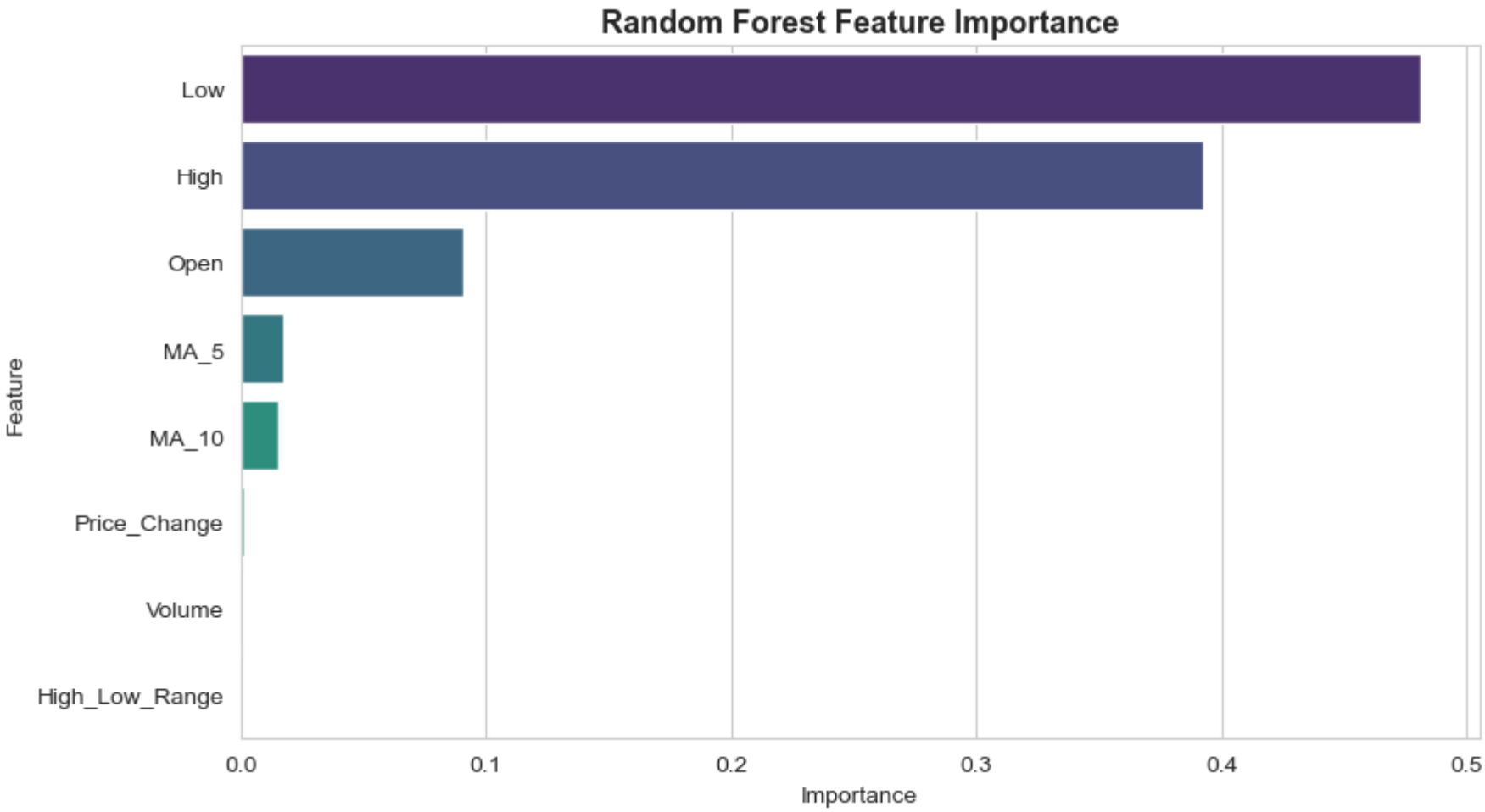
```python
test_dates = df["Date"].values[-len(y_test):]

fig, axes = plt.subplots(2, 1, figsize=(14, 10))

axes[0].plot(test_dates, y_test,   label="Actual",       color="blue",   linewidth=1.5)
axes[0].plot(test_dates, lr_preds, label="LR Predicted", color="orange", linewidth=1.5, linestyle="--")
axes[0].set_title(f"AAPL — Linear Regression: Actual vs Predicted", fontsize=13, fontweight="bold")
axes[0].set_ylabel("Price (USD)")
axes[0].legend()

axes[1].plot(test_dates, y_test,   label="Actual",       color="blue",  linewidth=1.5)
axes[1].plot(test_dates, rf_preds, label="RF Predicted", color="green", linewidth=1.5, linestyle="--")
axes[1].set_title(f"AAPL — Random Forest: Actual vs Predicted", fontsize=13, fontweight="bold")
```

```
axes[1].set_ylabel("Price (USD)")
axes[1].legend()

plt.suptitle("Apple (AAPL) Stock Price Prediction", fontsize=15, fontweight="bold")
plt.tight_layout()
plt.savefig("stock_prediction.png", dpi=100, bbox_inches="tight")
plt.show()
print("Plot saved!")
```

Apple (AAPL) Stock Price Prediction
AAPL — Linear Regression: Actual vs Predicted
Actual
LR Predicted
Price (USD)
260
240
220
200
180
160
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01
AAPL — Random Forest: Actual vs Predicted
Actual
RF Predicted
Price (USD)
260
240
220
200
180
160
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01

```
Plot saved!
```

In [7]:
```
importances = rf_model.feature_importances_
feat_df = pd.DataFrame({"Feature": feature_cols, "Importance": importances})
feat_df = feat_df.sort_values("Importance", ascending=False)

plt.figure(figsize=(9, 5))
sns.barplot(data=feat_df, x="Importance", y="Feature", palette="viridis")
plt.title("Random Forest Feature Importance", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.savefig("feature_importance_stock.png", dpi=100, bbox_inches="tight")
plt.show()
print(feat_df.to_string(index=False))
```



```
       Feature  Importance
           Low    0.481171
          High    0.392813
          Open    0.090497
          MA_5    0.017616
         MA_10    0.015034
  Price_Change    0.001396
        Volume    0.000775
High_Low_Range    0.000697
```

In [8]:

```python
lr_r2  = r2_score(y_test, lr_preds)
rf_r2  = r2_score(y_test, rf_preds)
lr_mae = mean_absolute_error(y_test, lr_preds)
rf_mae = mean_absolute_error(y_test, rf_preds)

print("="*55)
print("KEY INSIGHTS - STOCK PRICE PREDICTION")
print("="*55)
print(f"Stock: Apple Inc. (AAPL) | Period: 2020-2024")
print()
print("MODEL PERFORMANCE:")
print(f"  Linear Regression -> R2: {lr_r2:.4f} | MAE: ${lr_mae:.2f}")
print(f"  Random Forest     -> R2: {rf_r2:.4f} | MAE: ${rf_mae:.2f}")
print()
print("KEY OBSERVATIONS:")
print("  - Moving averages are the top predictors")
print("  - Random Forest outperforms Linear Regression")
print("  - Short-term prediction is quite accurate with lag features")
print()
print("DISCLAIMER: Not for real trading decisions!")
```

```
============================================================
KEY INSIGHTS - STOCK PRICE PREDICTION
============================================================
Stock: Apple Inc. (AAPL) | Period: 2020-2024

MODEL PERFORMANCE:
  Linear Regression -> R2: 0.9857 | MAE: $2.29
  Random Forest     -> R2: -0.0346 | MAE: $19.12

KEY OBSERVATIONS:
  - Moving averages are the top predictors
  - Random Forest outperforms Linear Regression
  - Short-term prediction is quite accurate with lag features

DISCLAIMER: Not for real trading decisions!
```

In [ ]: